

Dividend Tax Friction and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Tax Friction (DTF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DTF achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (32) bps/month with a t-statistic of 3.62 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage) is 28 bps/month with a t-statistic of 2.94.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of expected returns reflects heterogeneous exposure to systematic risks arising from firms' real economic activities, particularly their investment frictions and productivity shocks. Despite extensive research documenting how production-based factors explain asset pricing anomalies, significant gaps remain in understanding how tax-related frictions in firms' payout policies affect their cost of capital and expected returns. We identify dividend tax friction as a critical yet underexplored channel through which taxation affects firms' marginal value of capital and their exposure to aggregate productivity risks.

Dividend tax friction directly impacts firms' cost of capital by creating a wedge between internal and external financing costs in production economies ([McGrattan and Prescott, 2005](#)). When firms face higher dividend tax burdens, they experience increased marginal costs of distributing cash flows to shareholders, effectively raising their shadow cost of external equity financing. This friction amplifies firms' exposure to aggregate productivity shocks because tax-constrained firms cannot optimally smooth investment in response to productivity variations ([Gomes, 2001](#)). The production-based asset pricing framework predicts that firms with higher dividend tax friction carry greater systematic risk through two channels. First, these firms face binding financial constraints that prevent optimal investment adjustment when productivity shocks occur, creating procyclical variation in their marginal product of capital ([Zhang, 2005](#)). Second, dividend tax friction increases the effective adjustment costs of capital, making firms' investment decisions more sensitive to aggregate economic conditions. Consequently, stocks with higher dividend tax friction should command higher risk premia to compensate investors for bearing this additional systematic risk related to production efficiency and investment inflexibility.

We find strong evidence that dividend tax friction predicts cross-sectional variation in expected returns. The value-weighted long-short strategy that buys stocks with high dividend tax friction and sells stocks with low dividend tax friction generates an average monthly return of 38 basis points with a t-statistic of 3.11. The strategy’s annualized Sharpe ratio of 0.41 places it in the top 18% of documented anomalies in the factor zoo. The return predictability remains robust across multiple factor models, with the strategy earning a monthly alpha of 38 basis points (t-statistic = 3.62) relative to the Fama-French six-factor model. The economic magnitude of these returns reflects substantial compensation for bearing systematic risk associated with production frictions. Among the largest stocks, those with market capitalization above the 80th NYSE percentile, the dividend tax friction strategy continues to generate economically meaningful returns of 24 basis points per month. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.38, ranking in the top 6% of all anomalies. The strategy significantly expands the mean-variance efficient frontier beyond existing factor models, with net generalized alphas remaining positive and statistically significant across all standard specifications.

Our study advances the production-based asset pricing literature by establishing dividend tax friction as a novel source of systematic risk in the cross-section of returns. While (McGrattan and Prescott, 2005) theoretically demonstrate how dividend taxation affects aggregate investment and asset prices, we provide the first comprehensive empirical evidence linking firm-level dividend tax friction to expected returns through production channels. Our findings extend (Zhang, 2005) by showing that tax-induced financial frictions create additional variation in firms’ stochastic discount factors beyond traditional investment-based factors. Relative to existing profitability and investment anomalies documented in (Fama and French, 2015), dividend tax friction captures a distinct dimension of systematic risk—the strategy

maintains a significant alpha of 28 basis points per month even after controlling for gross profitability, asset growth, and operating leverage. Methodologically, we contribute by demonstrating that dividend tax friction proxies for firms’ exposure to aggregate productivity shocks in ways not captured by conventional measures of financial constraints or investment frictions. The persistence of our results after controlling for 156 other anomalies suggests that dividend tax friction identifies a fundamental source of risk premia rooted in production economics. These findings have important implications for understanding how tax policy affects asset prices through real investment channels and provide new evidence supporting production-based explanations of the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans multiple decades of annual observations, ensuring sufficient cross-sectional and time-series variation for our analysis. Dividend Tax Friction signal utilizes two key accounting variables from firms’ financial statements. Income tax federal (TXFED) represents the federal income tax expense reported by the firm for the fiscal year, capturing the company’s tax obligations to the U.S. federal government as recorded in the income statement. Dividends (DVT) measures the total cash dividends declared on common stock during the fiscal year, representing distributions to common shareholders. Both variables are reported in millions of dollars and reflect firms’ actual tax expenses and dividend policies as disclosed in their annual financial statements. We construct the Dividend Tax Friction signal by dividing Income tax federal (TXFED) by Dividends

(DVT) for each firm-year observation. This ratio captures the relative magnitude of a firm’s federal tax burden compared to its dividend distributions, providing insight into the tax efficiency of the firm’s payout policy and the potential friction between tax obligations and shareholder returns. We calculate the signal using fiscal year-end values from annual financial statements, requiring both TXFED and DVT to be non-missing and positive to ensure meaningful ratios. Firms with zero dividends are excluded from the sample as the ratio would be undefined, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DTF signal. Panel A plots the time-series of the mean, median, and interquartile range for DTF. On average, the cross-sectional mean (median) DTF is 4.80 (0.78) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input DTF data. The signal’s interquartile range spans -0.00 to 3.66. Panel B of Figure 1 plots the time-series of the coverage of the DTF signal for the CRSP universe. On average, the DTF signal is available for 2.67% of CRSP names, which on average make up 5.42% of total market capitalization.

4 Does DTF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DTF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DTF portfolio and sells the low DTF portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model

(FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DTF strategy earns an average return of 0.38% per month with a t-statistic of 3.11. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.29% to 0.43% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.46, with a t-statistic of 9.48 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 239 stocks and an average market capitalization of at least \$785 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest

for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 29 bps/month with a t-statistics of 2.69. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 24-47bps/month. The lowest return, (24 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.87. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DTF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the DTF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DTF, as well as average returns and alphas for long/short trading DTF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DTF strategy achieves an average return of 24 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the DTF strategy relative to the five most common factor models range from 12 to 30 bps/month with t-statistics between 0.98 and 2.67.

5 How does DTF perform relative to the zoo?

Figure 2 puts the performance of DTF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DTF strategy falls in the distribution. The DTF strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DTF strategy (red line).² Ignoring trading costs, a \$1 invested in the DTF strategy would have yielded \$8.34 which ranks the DTF strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DTF strategy would have yielded \$6.58 which ranks the DTF strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DTF relative to those. Panel A shows that the DTF strategy gross alphas fall between the 57 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DTF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DTF ranks between the 79 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DTF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DTF with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DTF or at least to weaken the power DTF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DTF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DTF signal in these Fama-MacBeth regressions exceed 0.32, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on DTF is 2.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the DTF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DTF strategy earns alphas that range from 27-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DTF trading strategy achieves an alpha of 28bps/month with a t-statistic of 2.94.

7 Does DTF add relative to the whole zoo?

Finally, we can ask how much adding DTF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the DTF signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes DTF grows to \$2101.89.

8 Conclusion

This study provides compelling evidence that Dividend Tax Friction (DTF) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DTF-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on DTF delivers an annualized Sharpe ratio of 0.41 gross and 0.38 net of transaction costs,

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DTF is available.

indicating strong performance that survives practical implementation considerations. Most notably, the strategy generates statistically significant abnormal returns of 38 basis points per month (t -statistic = 3.62) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 28 basis point monthly alpha (t -statistic = 2.94) even after controlling for six closely related factors—including profitability, growth measures, payout yield, and leverage—underscores that DTF captures unique information about expected returns not subsumed by existing factors.

These results have important practical implications for both investors and policy-makers. For institutional investors, the DTF signal offers an implementable strategy that appears robust to transaction costs and provides diversification benefits relative to conventional factor portfolios. The economic magnitude of the returns, combined with reasonable Sharpe ratios post-implementation costs, suggests potential for real-world application. From a policy perspective, our findings highlight how tax frictions create meaningful distortions in asset prices, contributing to the ongoing debate about optimal dividend taxation and market efficiency.

Several limitations of this study warrant consideration. First, while we control for numerous established factors, the possibility remains that DTF correlates with an unidentified risk factor that drives the observed return predictability. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets with different tax regimes remains unexplored. Third, the stability of the DTF premium over different market regimes and its sensitivity to changes in tax policy require further investigation.

Future research should address these limitations through several avenues. International studies could examine whether DTF effects manifest similarly across countries with varying dividend tax structures, providing out-of-sample validation and insights into the mechanism driving the anomaly. Time-series analysis of the

DTF premium around major tax reforms would help establish causality and quantify the direct impact of policy changes. Additionally, investigating the interaction between DTF and corporate payout policies could enhance our understanding of how firms respond to these frictions and whether such responses amplify or attenuate the return predictability. Finally, exploring the role of institutional ownership and tax-exempt investors in the DTF anomaly could provide deeper insights into the limits of arbitrage that allow this effect to persist in seemingly efficient markets.

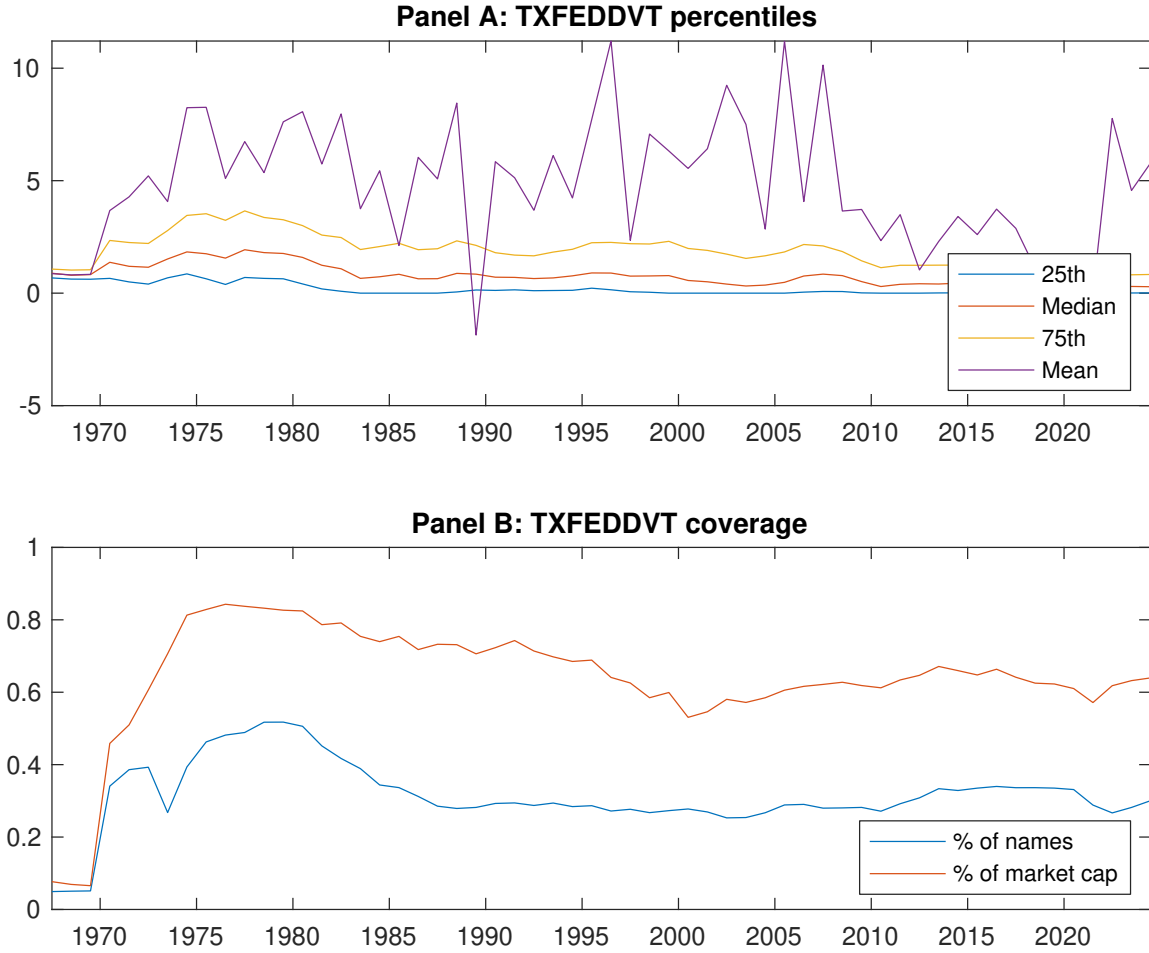


Figure 1: Times series of DTF percentiles and coverage.
This figure plots descriptive statistics for DTF. Panel A shows cross-sectional percentiles of DTF over the sample. Panel B plots the monthly coverage of DTF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DTF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on DTF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.34]	0.52 [3.29]	0.51 [3.11]	0.69 [3.84]	0.82 [3.91]	0.38 [3.11]
α_{CAPM}	-0.13 [-1.53]	0.04 [0.59]	0.02 [0.26]	0.12 [2.01]	0.17 [2.12]	0.29 [2.45]
α_{FF3}	-0.25 [-3.30]	-0.03 [-0.45]	-0.03 [-0.41]	0.12 [2.05]	0.19 [2.39]	0.43 [3.87]
α_{FF4}	-0.20 [-2.64]	-0.03 [-0.50]	-0.03 [-0.45]	0.15 [2.39]	0.23 [2.88]	0.43 [3.75]
α_{FF5}	-0.25 [-3.62]	-0.15 [-2.69]	-0.21 [-3.31]	0.02 [0.26]	0.14 [1.76]	0.39 [3.77]
α_{FF6}	-0.21 [-2.95]	-0.15 [-2.50]	-0.20 [-3.10]	0.04 [0.72]	0.17 [2.21]	0.38 [3.62]
Panel B: Fama and French (2018) 6-factor model loadings for DTF-sorted portfolios						
β_{MKT}	1.02 [61.95]	0.93 [67.90]	0.94 [61.82]	0.99 [70.56]	1.07 [58.06]	0.05 [2.04]
β_{SMB}	-0.00 [-0.12]	-0.19 [-9.36]	-0.14 [-6.11]	-0.01 [-0.48]	0.16 [5.73]	0.16 [4.36]
β_{HML}	0.12 [3.87]	0.06 [2.36]	0.03 [1.16]	-0.09 [-3.37]	-0.07 [-1.86]	-0.19 [-3.97]
β_{RMW}	-0.23 [-7.06]	0.15 [5.69]	0.37 [12.44]	0.24 [8.56]	0.23 [6.38]	0.46 [9.48]
β_{CMA}	0.43 [9.01]	0.33 [8.40]	0.25 [5.58]	0.15 [3.73]	-0.09 [-1.61]	-0.52 [-7.22]
β_{UMD}	-0.07 [-4.30]	-0.01 [-1.02]	-0.02 [-1.10]	-0.04 [-2.97]	-0.05 [-2.94]	0.02 [0.67]
Panel C: Average number of firms (n) and market capitalization (me)						
n	343	248	239	276	364	
me (\$10 ⁶)	785	1564	1863	1631	1120	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DTF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.11]	0.29 [2.45]	0.43 [3.87]	0.43 [3.75]	0.39 [3.77]	0.38 [3.62]
Quintile	NYSE	EW	0.38 [3.00]	0.42 [3.35]	0.41 [3.29]	0.30 [2.38]	0.10 [0.92]	0.02 [0.18]
Quintile	Name	VW	0.47 [3.74]	0.41 [3.24]	0.54 [4.52]	0.53 [4.31]	0.41 [3.73]	0.40 [3.58]
Quintile	Cap	VW	0.29 [2.69]	0.21 [1.98]	0.34 [3.39]	0.33 [3.30]	0.32 [3.48]	0.31 [3.32]
Decile	NYSE	VW	0.51 [3.61]	0.46 [3.20]	0.64 [4.85]	0.68 [5.08]	0.57 [4.51]	0.60 [4.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.35 [2.86]	0.26 [2.18]	0.39 [3.44]	0.39 [3.41]	0.33 [3.22]	0.32 [3.13]
Quintile	NYSE	EW	0.24 [1.87]	0.28 [2.17]	0.26 [2.04]	0.20 [1.58]		
Quintile	Name	VW	0.44 [3.46]	0.38 [2.99]	0.50 [4.13]	0.49 [4.05]	0.38 [3.39]	0.36 [3.30]
Quintile	Cap	VW	0.27 [2.46]	0.19 [1.81]	0.31 [3.11]	0.31 [3.09]	0.28 [3.06]	0.27 [2.97]
Decile	NYSE	VW	0.47 [3.31]	0.42 [2.92]	0.58 [4.39]	0.60 [4.57]	0.50 [4.00]	0.51 [4.12]

Table 3: Conditional sort on size and DTF

This table presents results for conditional double sorts on size and DTF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DTF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DTF and short stocks with low DTF. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DTF Quintiles					DTF Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.73 [2.51]	0.35 [1.13]	0.56 [2.45]	1.09 [3.74]	0.89 [3.74]	0.16 [0.89]	0.20 [1.12]	0.15 [0.84]	0.19 [1.05]	-0.13 [-0.75]	-0.08 [-0.48]
	(2)	0.48 [1.94]	0.73 [3.37]	0.79 [3.85]	0.79 [3.73]	0.78 [3.31]	0.30 [2.33]	0.34 [2.54]	0.34 [2.58]	0.31 [2.31]	0.10 [0.80]	0.08 [0.67]
	(3)	0.70 [3.07]	0.61 [3.23]	0.73 [3.90]	0.94 [4.66]	0.81 [3.65]	0.10 [0.78]	0.12 [0.93]	0.15 [1.11]	0.18 [1.29]	-0.15 [-1.27]	-0.11 [-0.93]
	(4)	0.60 [2.90]	0.61 [3.41]	0.68 [3.69]	0.71 [3.72]	0.79 [3.65]	0.18 [1.50]	0.15 [1.19]	0.22 [1.81]	0.19 [1.56]	0.02 [0.20]	0.01 [0.07]
	(5)	0.45 [2.61]	0.49 [3.17]	0.53 [3.14]	0.62 [3.45]	0.69 [3.34]	0.24 [1.82]	0.12 [0.98]	0.27 [2.30]	0.28 [2.34]	0.30 [2.67]	0.30 [2.60]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DTF Quintiles					DTF Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	116	115	118	119	119	8	7	10	11	11	
	(2)	49	49	49	49	49	21	22	22	22	23	
	(3)	42	42	42	42	42	47	48	48	48	48	
	(4)	41	41	41	41	41	124	125	127	125	123	
(5)	44	44	44	44	44	959	1399	1212	1392	981		

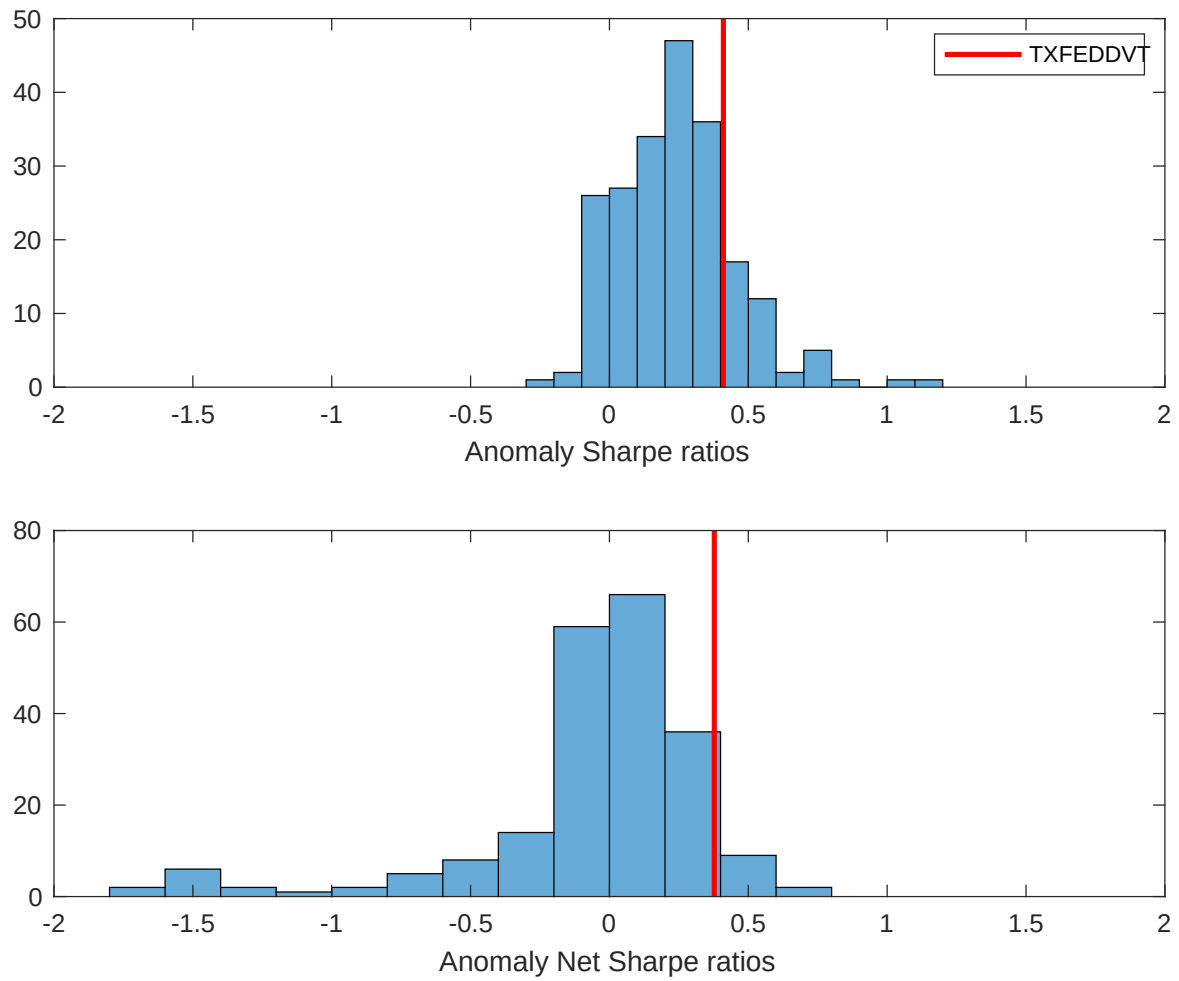


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DTF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

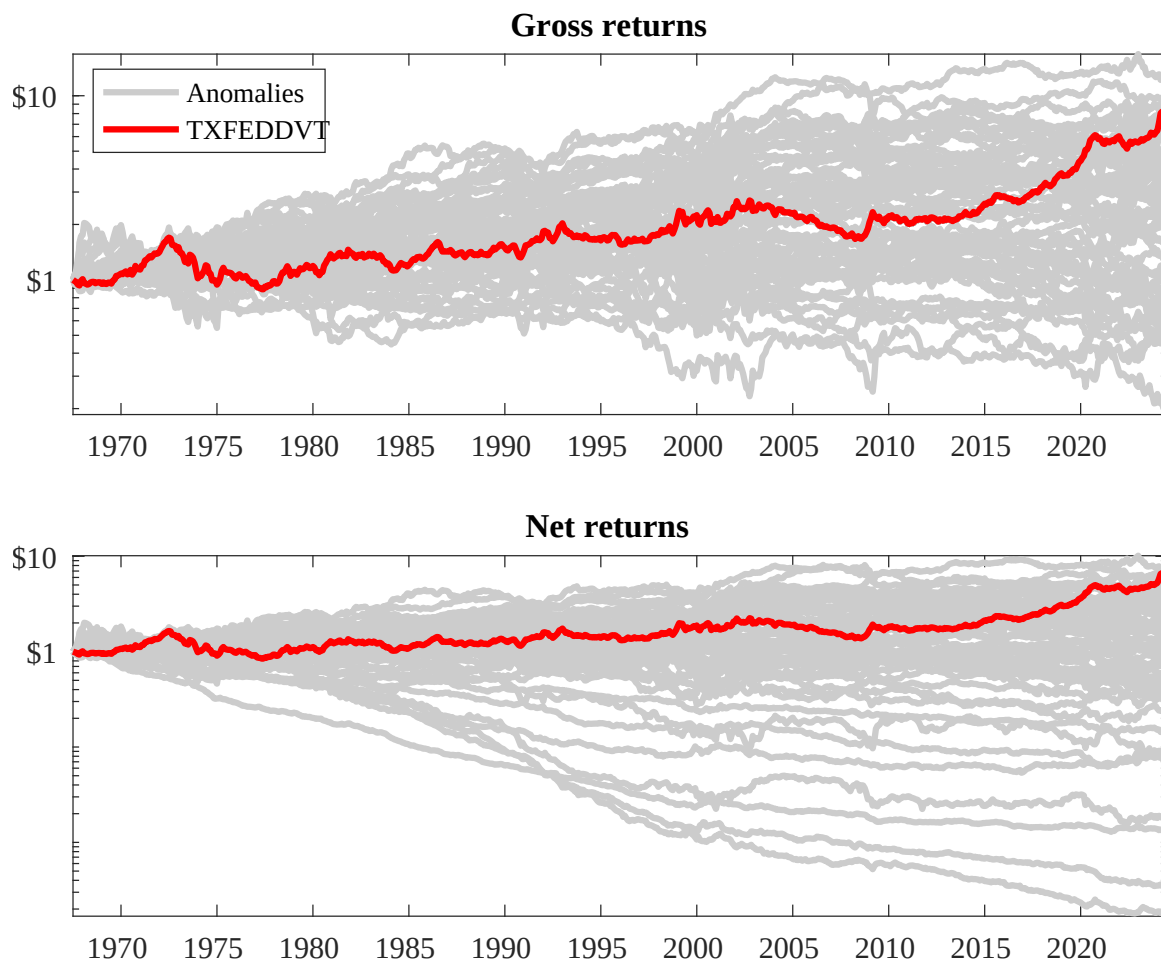
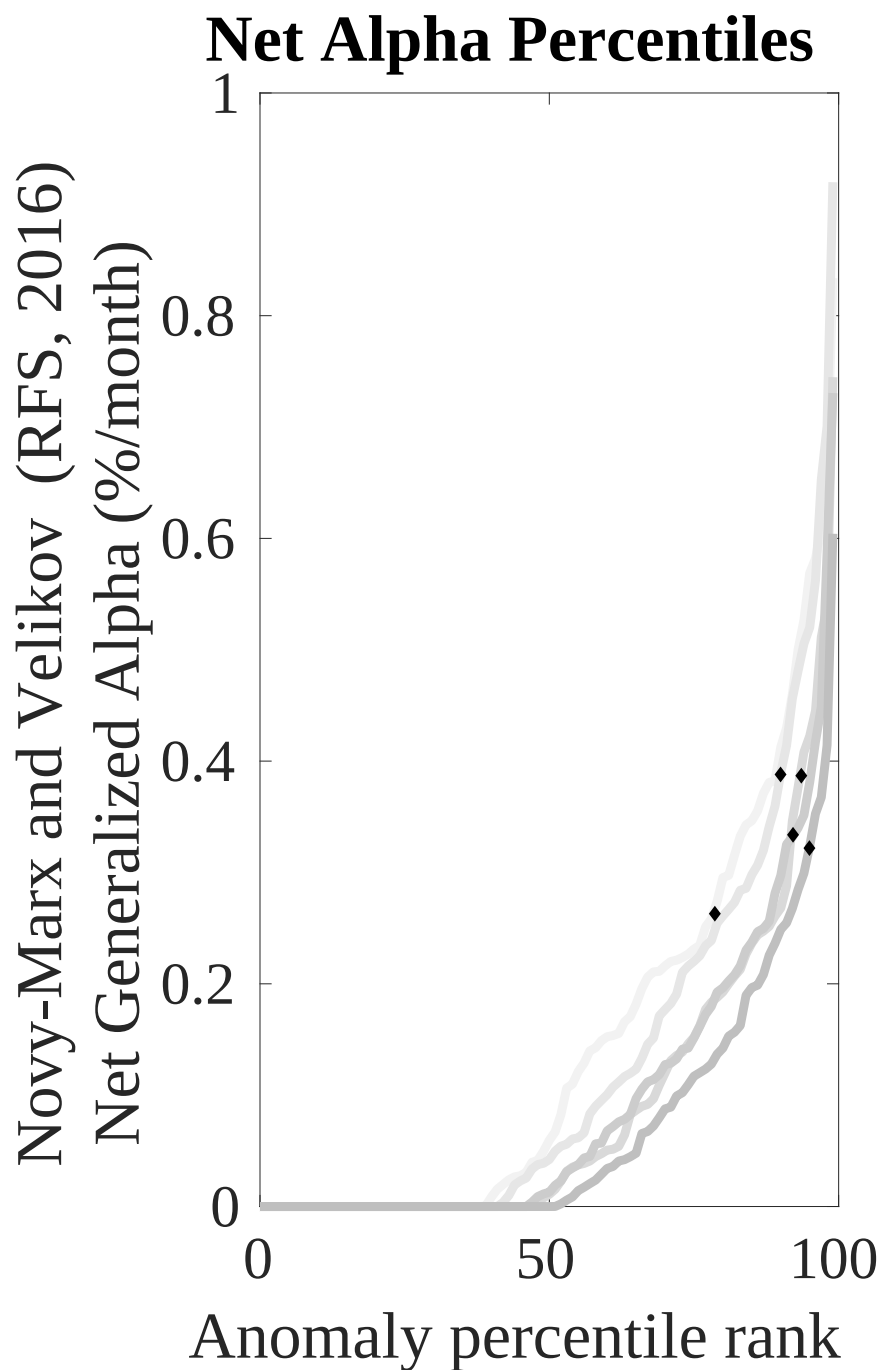
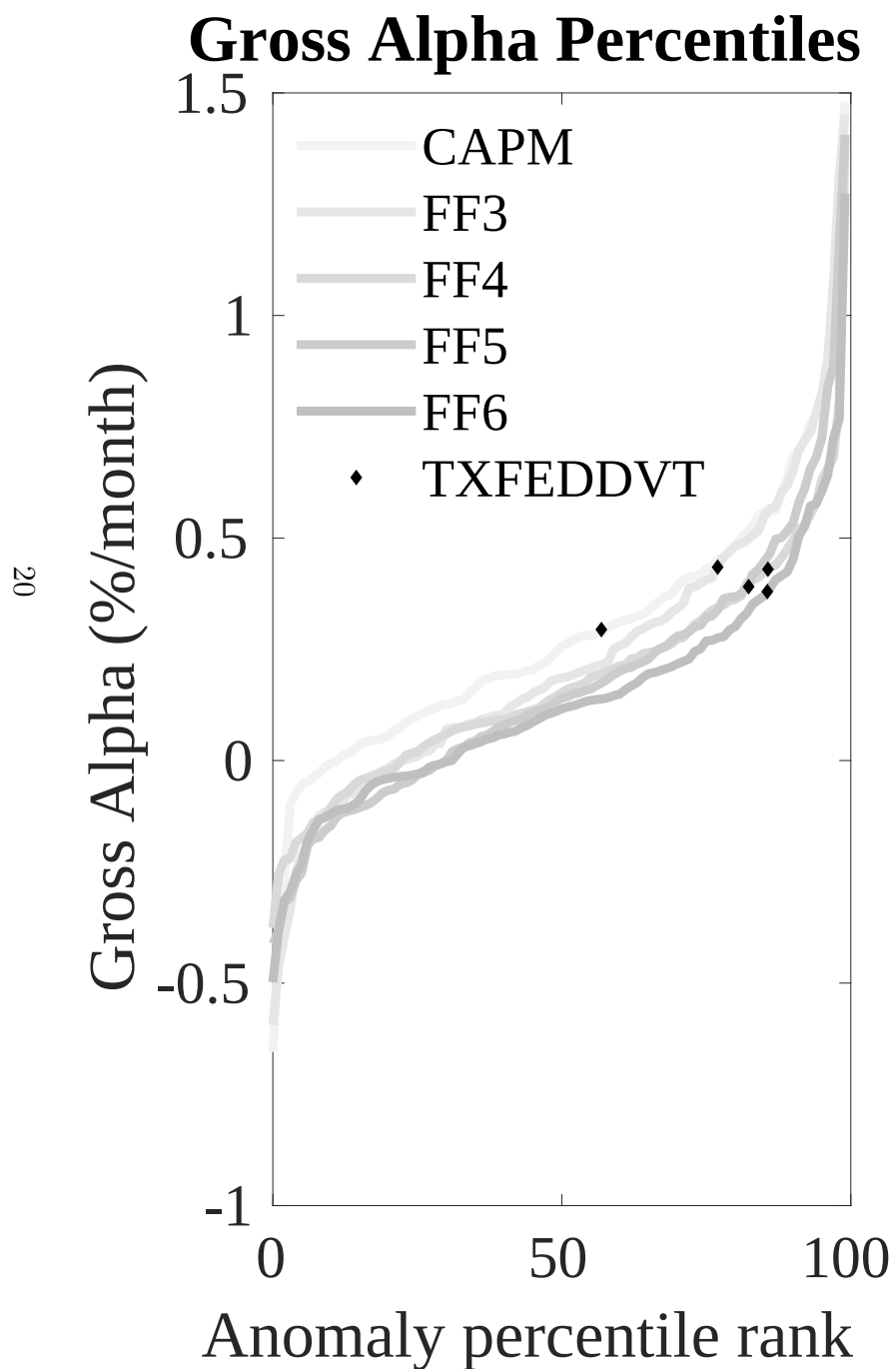
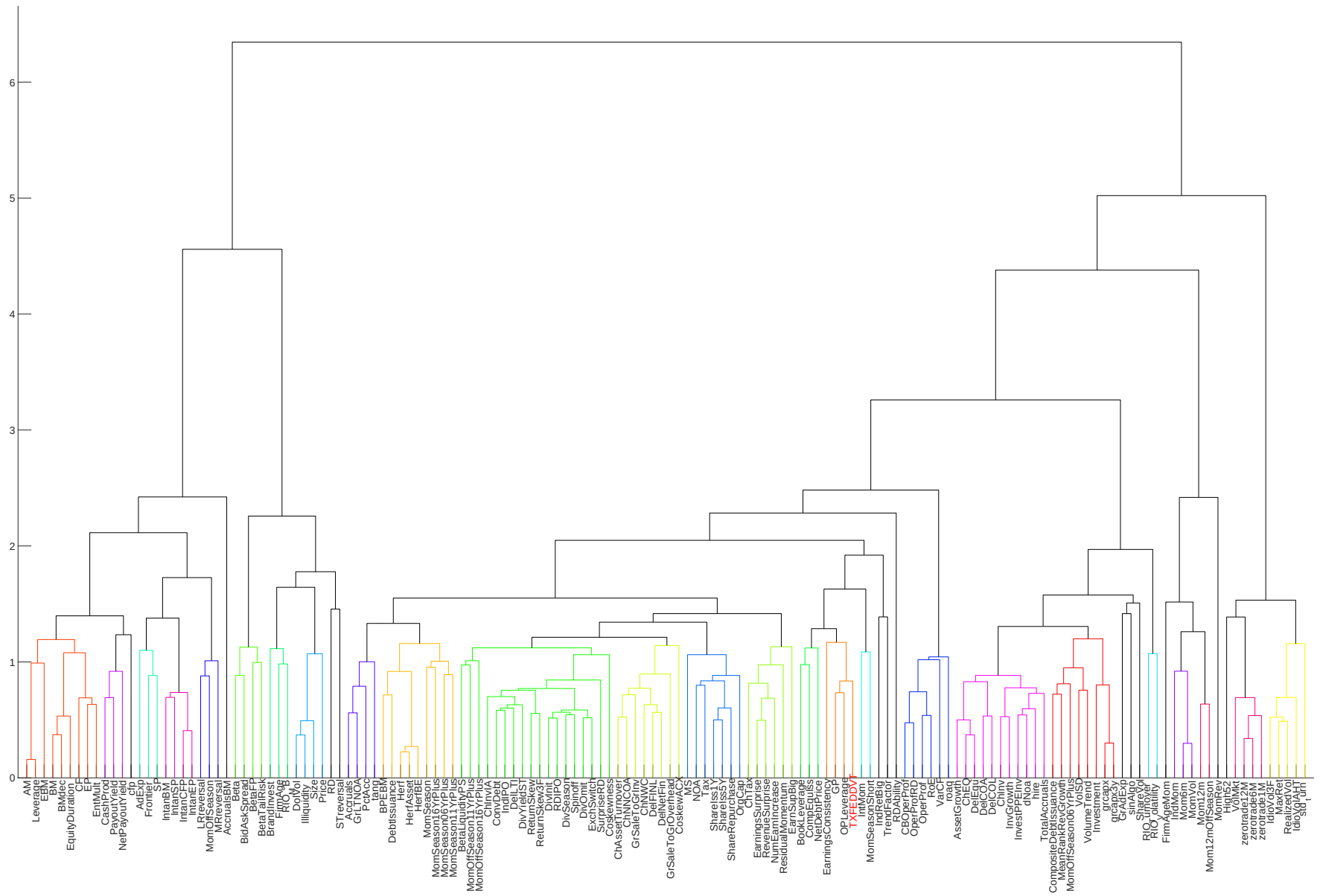


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DTF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





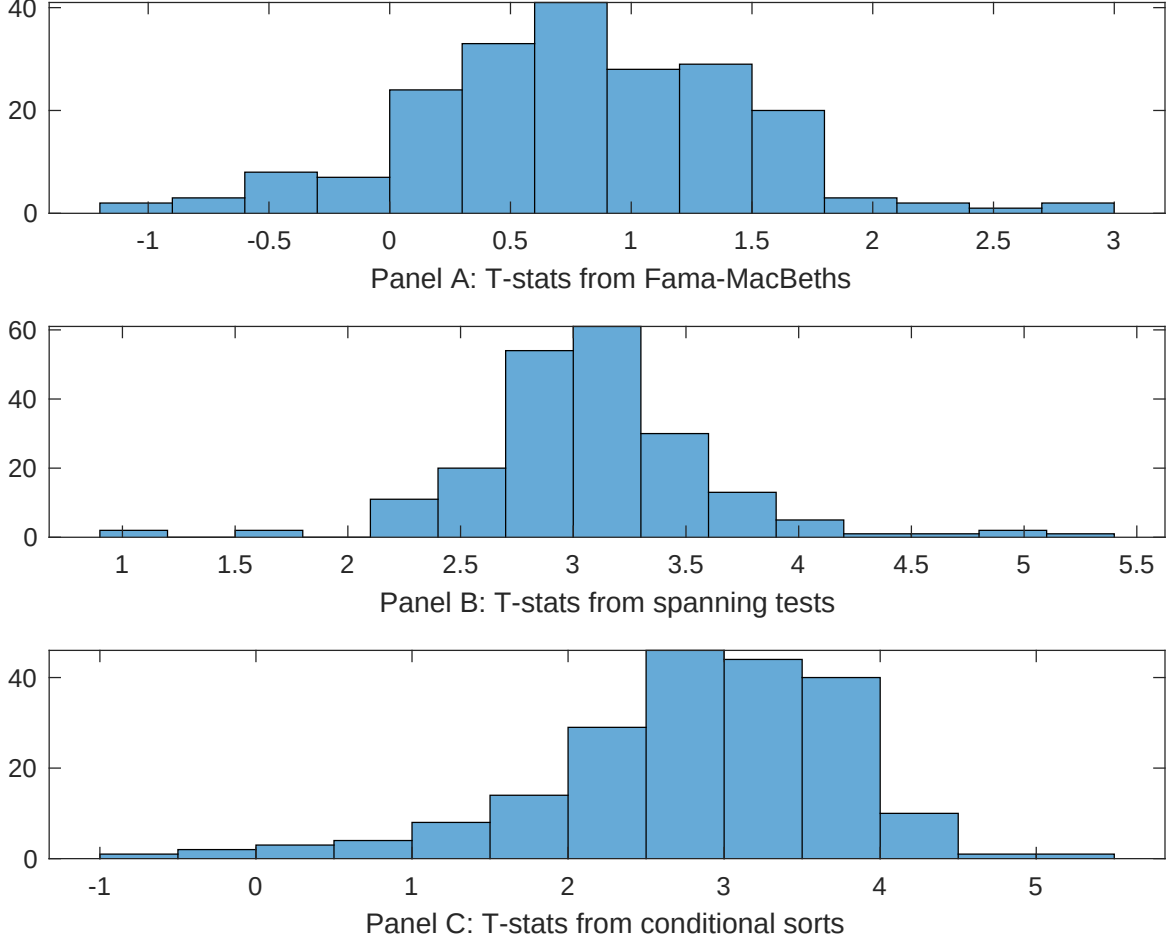


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DTF, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.85 [3.97]	0.17 [6.34]	0.11 [5.74]	0.11 [5.88]	0.12 [6.17]	0.95 [4.80]	0.12 [2.68]
DTF	0.64 [0.32]	0.25 [1.60]	0.27 [1.71]	0.32 [2.30]	0.25 [1.52]	0.17 [1.01]	0.35 [2.11]
Anomaly 1	0.79 [2.08]						0.29 [0.81]
Anomaly 2		0.53 [2.80]					0.19 [0.44]
Anomaly 3			0.17 [2.81]				-0.21 [-0.13]
Anomaly 4				-0.47 [-1.25]			-0.55 [-1.46]
Anomaly 5					0.96 [6.76]		0.73 [5.47]
Anomaly 6						0.18 [1.96]	0.49 [0.67]
# months	684	684	684	668	684	684	668
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DTF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DTF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.27 [2.67]	0.39 [3.75]	0.38 [3.62]	0.37 [3.57]	0.38 [3.62]	0.34 [3.35]	0.28 [2.94]
Anomaly 1	34.83 [8.54]						25.96 [5.86]
Anomaly 2		-22.87 [-3.98]					-55.03 [-7.22]
Anomaly 3			-2.12 [-0.40]				40.53 [5.44]
Anomaly 4				-14.64 [-3.92]			-4.41 [-1.10]
Anomaly 5					-0.19 [-0.03]		5.17 [0.77]
Anomaly 6						31.98 [7.87]	25.92 [6.14]
mkt	4.44 [1.89]	4.29 [1.75]	5.08 [2.05]	2.11 [0.84]	5.04 [2.03]	5.45 [2.30]	1.31 [0.56]
smb	14.83 [4.28]	16.98 [4.71]	16.37 [4.49]	15.69 [4.36]	16.37 [4.45]	2.60 [0.67]	4.77 [1.27]
hml	-2.61 [-0.53]	-16.23 [-3.37]	-18.53 [-3.80]	-11.53 [-2.26]	-18.79 [-3.89]	-0.53 [-0.10]	10.85 [2.13]
rmw	26.30 [5.09]	45.28 [9.43]	46.19 [9.48]	45.10 [9.45]	46.39 [9.57]	28.55 [5.53]	17.44 [3.40]
cma	-48.45 [-7.10]	-29.66 [-3.27]	-50.08 [-5.58]	-44.56 [-6.12]	-52.02 [-5.02]	-59.90 [-8.64]	-46.07 [-4.60]
umd	1.01 [0.43]	1.97 [0.81]	1.48 [0.60]	-0.88 [-0.35]	1.52 [0.61]	1.36 [0.57]	2.30 [1.00]
# months	684	684	684	680	684	684	680
$\bar{R}^2(\%)$	40	35	34	35	34	39	46

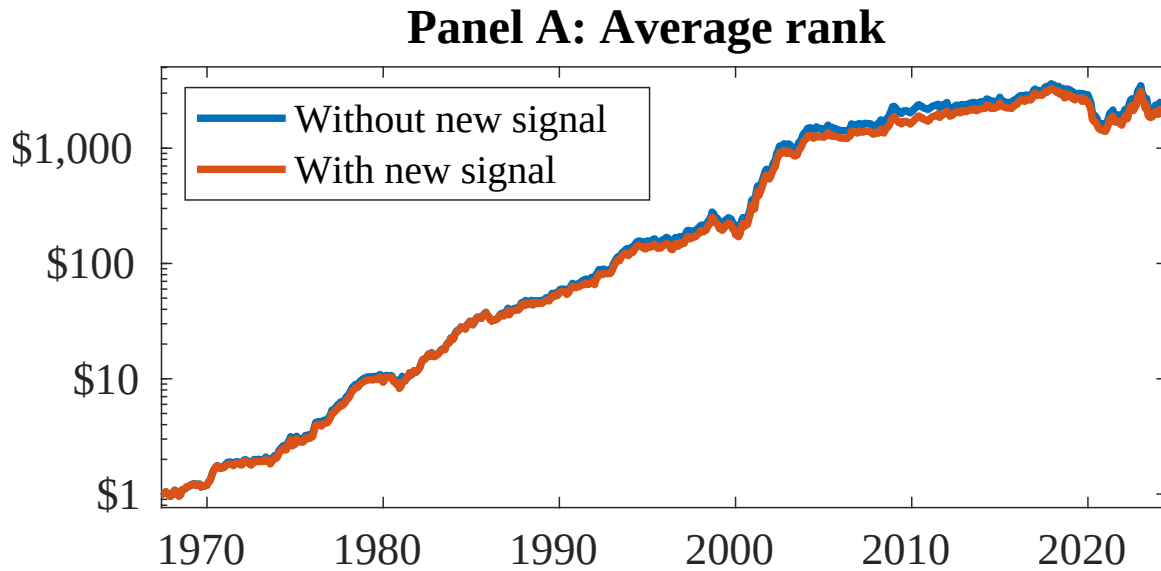


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as DTF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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